

Topic 16: The Laplace Transform (The Time-to-Frequency Machine)

Professor Leonard Master Series

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Algebraic Time Machine

Professor Leonard introduces the Laplace Transform with wonder:

"The Laplace Transform is a magical time machine. In the time domain (t), calculus is brutal: derivatives, integrals, initial conditions tangled together. The Laplace Transform beams your differential equation into the complex frequency domain (s), where derivatives turn into simple multiplication by s ! You solve the equation using basic grade-school algebra, and then take the Inverse Laplace Transform to beam the solution back into real time!"

$$\mathcal{L}f(t) = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

2. Fundamental Transform Dictionary & Operational Theorems

1. $\mathcal{L}1 = \frac{1}{s}$
2. $\mathcal{L}t^n = \frac{n!}{s^{n+1}}$
3. $\mathcal{L}e^{at} = \frac{1}{s-a}$
4. $\mathcal{L}\sin kt = \frac{k}{s^2+k^2}$, $\mathcal{L}\cos kt = \frac{s}{s^2+k^2}$
5. **Derivatives Theorem:**

$$\mathcal{L}y' = sY(s) - y(0)$$

$$\mathcal{L}y'' = s^2Y(s) - sy(0) - y'(0)$$

6. **First Shifting Theorem:** $\mathcal{L}e^{at}f(t) = F(s-a)$
 7. **Second Shifting Theorem (Heaviside Step):** $\mathcal{L}f(t-a)\mathcal{U}(t-a) = e^{-as}F(s)$
 8. **Dirac Delta Impulse:** $\mathcal{L}\delta(t-t_0) = e^{-st_0}$
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3. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 16.1: Solving a 2nd-Order IVP with Repeated Root

Statement: Solve $y'' - 4y' + 4y = e^{2t}$ subject to $y(0) = 0, y'(0) = 1$.

Step-by-Step Whiteboard Solution:

- **Step 1: Take Laplace Transform of both sides:**

$$\mathcal{L}y'' - 4\mathcal{L}y' + 4\mathcal{L}y = \mathcal{L}e^{2t}$$

- **Step 2: Substitute Derivative Formulas & Initial Conditions:**

$$(s^2Y - sy(0) - y'(0)) - 4(sY - y(0)) + 4Y = \frac{1}{s-2}$$

$$(s^2Y - 0 - 1) - 4(sY - 0) + 4Y = \frac{1}{s-2}$$

$$(s^2 - 4s + 4)Y - 1 = \frac{1}{s-2}$$

- **Step 3: Solve algebraically for $Y(s)$:**

$$(s-2)^2Y = 1 + \frac{1}{s-2} = \frac{s-2+1}{s-2} = \frac{s-1}{s-2}$$

$$Y(s) = \frac{1}{(s-2)^2} + \frac{1}{(s-2)^3}$$

- **Step 4: Take the Inverse Laplace Transform using First Shifting Theorem:**

◦ For $\frac{1}{(s-2)^2}$: Since $\mathcal{L}t = \frac{1}{s^2}$, shifting by $a = 2$ gives $\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\} = te^{2t}$.

◦ For $\frac{1}{(s-2)^3}$: Since $\mathcal{L}t^2 = \frac{2!}{s^3} = \frac{2}{s^3}$, $\frac{1}{(s-2)^3} = \frac{1}{2} \cdot \frac{2!}{(s-2)^3} \implies \frac{1}{2}t^2e^{2t}$.

- **Step 5: Assemble Final Time-Domain Solution:**

$$y(t) = te^{2t} + \frac{1}{2}t^2e^{2t} = te^{2t} \left(1 + \frac{1}{2}t\right)$$

Problem 16.2: Completing the Square for Underdamped Frequency

Statement: Find $\mathcal{L}^{-1}\left\{\frac{2s+5}{s^2+4s+13}\right\}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Complete the Square in the Denominator:**

$$s^2 + 4s + 13 = (s+2)^2 + 9 = (s+2)^2 + 3^2$$

This indicates an exponential shift by $a = -2$ and oscillation frequency $k = 3$.

- **Step 2: Express Numerator in terms of $(s+2)$:**

$$2s+5 = 2(s+2) - 4 + 5 = 2(s+2) + 1$$

- **Step 3: Split into Cosine and Sine components:**

$$\frac{2(s+2)+1}{(s+2)^2+3^2} = 2\frac{s+2}{(s+2)^2+3^2} + \frac{1}{3}\frac{3}{(s+2)^2+3^2}$$

- **Step 4: Apply Inverse Laplace Transform:**

$$f(t) = 2e^{-2t} \cos 3t + \frac{1}{3}e^{-2t} \sin 3t = e^{-2t} \left(2 \cos 3t + \frac{1}{3} \sin 3t \right)$$

Problem 16.3: Discontinuous Forcing & The Unit Step Function

Statement: Solve $y' + 2y = g(t)$, $y(0) = 0$ where $g(t) = \begin{cases} 1, & 0 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Express $g(t)$ using Heaviside Step Functions:**

$$g(t) = 1 - \mathcal{U}(t - 3)$$

- **Step 2: Take Laplace Transform:**

$$(sY - 0) + 2Y = \frac{1}{s} - \frac{e^{-3s}}{s}$$

$$(s+2)Y = \frac{1 - e^{-3s}}{s} \implies Y(s) = (1 - e^{-3s}) \frac{1}{s(s+2)}$$

- **Step 3: Partial Fractions on $\frac{1}{s(s+2)}$:**

$$\frac{1}{s(s+2)} = \frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right)$$

$$H(s) = \frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right) \implies h(t) = \frac{1}{2}(1 - e^{-2t})$$

- **Step 4: Apply Second Shifting Theorem to $e^{-3s}H(s)$:**

$$y(t) = h(t) - h(t-3)\mathcal{U}(t-3)$$

$$y(t) = \frac{1}{2}(1 - e^{-2t}) - \frac{1}{2}(1 - e^{-2(t-3)})\mathcal{U}(t-3)$$

Problem 16.4: Dirac Delta Hammer Impulse

Statement: Solve $y'' + y = \delta(t - \pi)$, $y(0) = 0$, $y'(0) = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Take Laplace Transform:**

$$(s^2 + 1)Y = e^{-\pi s} \implies Y(s) = e^{-\pi s} \frac{1}{s^2 + 1}$$

- **Step 2: Invert using Second Shifting Theorem:** Since $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$:

$$y(t) = \sin(t - \pi)\mathcal{U}(t - \pi) = -\sin(t)\mathcal{U}(t - \pi)$$

The system sits completely motionless until $t = \pi$, when the hammer strikes, initiating pure sinusoidal oscillation!

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting Initial Conditions in Derivative Formulas:** $\mathcal{L}y' = sY - y(0)$. If $y(0) \neq 0$, you must subtract it!
- **Trap 2: Unit Step Shift Mismatch:** In $\mathcal{L}f(t)\mathcal{U}(t - a)$, the function must be shifted as $f(t - a)$ before applying $e^{-as}F(s)$!